Bruno Magalhaes — High Performance Computing and Machine Learning

brunomaga@gmail.com | brunomaga | brunomaga | https://brunomaga.github.io | Lausanne, Switzerland
C, C++, CUDA, DeepSpeed, HPX, MPI, Python, PyTorch, shmem, Triton | Portuguese-Swiss | English, French, Portuguese, Spanish

Work Experience

present – Jul 2025 — **Principal Researcher for AI Systems and Networks**, Huawei Research, Zurich, Switzerland
- Started, grew and now leading the AI DC Networks team (7), driving research direction and technical implementation.
- Distributed inference with MegaKernels (CUDA, nvshmem) that fuse computation, communication and in-network compute.
- Applied to FFNs, attention, MoE and, recently, agentic AI; deployed in open-source DeepSeek and vLLM modules.
- Scaling distributed models via tensor, data, expert and context parallelism, and sharding of MoE, FFN and attention layers.
- Supervising collaborations with universities on projects, grants and internships.

May 2025 – Nov 2023 — **Senior Research Engineer for Large-scale AI**, Synthesia, Zurich, Switzerland
- Started, grew and led the ML performance team (4) for large-scale training and inference of diffusion and transformer models (text, audio, video): data and sequence parallelism (Ulysses, Ring attention), sharding, activation checkpointing, scaling laws.
- Model speedup via kernels fusion (Triton, CUDA, torch compile), quantisation (int8 Sage attention), and channels-last format.
- Created the DeepSpeed modules for distributed curriculum learning setup (PR 5129) and variable batch size and LR (PR 7104).

Oct 2023 – Sep 2019 — **AI Resident › Researcher › Senior Researcher**, Microsoft Research, Cambridge, UK
- as Sr. Researcher: Efficient inference of vision models on optical hardware with pipeline and model parallelism (C++). Large-scale ML training via data parallelism, sharding, gradient accumulation, activation checkpointing, IO offloading and distillation (DeepSpeed/ZeRO and Torch Distributed/RPC). Distributed Mixture of Experts. Likelihood estimators, information encoding and error correction (LDPC) for storage systems. Mentoring of junior members and PhD interns.
- as Researcher: Computer vision models for thousand-object classification on 3D glass at Project Silica. Presenter of talks on the topics of *CPU/GPU optimisation*, *distributed algorithms* and *AI SuperComputing*.
- as AI Resident: RNNs and Bayesian Optimisation on time series for load balancing of Exchange email servers on exabyte-scale databases. Graph Neural Nets for recommendations on a trillion-edge graph of meetings, documents, emails and users.

Aug 2019 – Mar 2011 — **HPC Engineer › PhD Candidate › Postdoc Researcher**, École Polytechnique Fédérale de Lausanne (EPFL), Switzerland
- as Postdoc: Supervision of PhD students and engineers. Scientific reviewer for *SuperComputing*, *IPDPS*, and *ISC* conferences.
- as PhD Candidate: Research, development (C, C++, HPX) and publication on the asynchronous variable-order variable-step simulation of spiking neural networks, distributed on Cray and SGI supercomputers (10K+ nodes) with a global address space (PGAS, RDMA, InfiniBand), distributed control objects, dynamic load balancing, vectorisation and cache optimisation.
- Teaching assistant (400h) for *Unsupervised and reinforcement learning*, *Project in neuroinformatics* and *In silico neuroscience*.
- as HPC Engineer: Research, development (C, C++, MPI, OpenMP) and publication of methods for parallel/distributed volumetric decomposition, load balancing, spatial indexing, sorting, IO, sparse matrix transpose and graph navigation that underlie efficient processing of neural networks on SGI and IBM Blue Gene/Q supercomputers with 16K+ compute nodes.

Feb 2011 – Sep 2009 — **Junior Architect for IT infrastructures**, Noble Group, London, New York, & São Paulo
- Design and configuration of Linux servers, CISCO networks, and backup/redundancy sites for physical trading of commodities.

Education

Jun 2019 — **PhD Computational Neuroscience**, École Polytechnique Fédérale de Lausanne (EPFL), Switzerland
- Thesis *Asynchronous Simulation of Neuronal Activity* nominated for the EPFL PhD excellence award and the IBM Research award for best thesis in computational sciences. Visiting researcher at Indiana University (US) during Summers of 2015, ’16 and ’17.

Sep 2009 — **MSc Advanced Computing**, Imperial College London, UK
- Grade: Merit. Thesis *GPU-enabled steady-state solution of large Markov models* awarded distinction and published at *NSMC’10*.

Jul 2007 — **BSc Systems Engineering and Computer Science**, University of Minho, Portugal
- Grade: A, top 10%. Exchange student at the University of Maribor, Slovenia, 2005/06. Intern at IBM. Part-time project at CERN.

Selected Publications

full list on Google Scholar

ongoing — Moeinc: fused Expert Parallelism with overlap of Matrix, Vector and In-Network Compute of AllToAll-Unique/Reduce
ongoing — Towards Efficient LLM Inference via Full-Stack Fusion of Computation, Communication, and In-Network Computing
2026 — Laser writing in glass for dense, fast and efficient archival data storage, Nature 650, pages 606–612
2024 — Project Silica: sustainable cloud archival storage in glass, Frontiers in Ultrafast Optics: Biomedical, Scientific, and Industrial Applications XXIV
2023 — Project Silica: Towards Sustainable Cloud Archival Storage in Glass, SOSP ’23: Proc. of the 29th Symposium on Operating Systems Principles
2020 — Fully-Asynchronous Fully-Implicit Variable-Order Variable-Timestep Simulation of Neural Networks, Proc. ICCS 2020, Amsterdam, Holland
2019 — Exploiting Implicit Flow Graph of System of ODEs to Accelerate the Simulation of Neural Networks, Proc. IPDPS 2019, Rio de Janeiro, Brazil
2015 — Reconstruction and Simulation of Neocortical Microcircuitry, Cell 163, 456–492.